Parameter-Efficient Adaptation of DINOv2 for Sub-Pixel Semantic Correspondence

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Abstract

Semantic correspondence seeks pixel-level matches between semantically related images, despite viewpoint, scale, and appearance variations. While the off-the-shelf features of visual foundation models like DINOv2 capture general semantic similarities, they lack the sub-pixel precision required for fine-grained matching and struggle significantly in cluttered scenes. To overcome these limitations, we propose a modular pipeline that enhances DINOv2 across both training and inference stages. For the training phase, we explore baseline feature adaptability through Linear Probing and comprehensively evaluate Parameter-Efficient Fine-Tuning (PEFT) strategies—specifically Low-Rank Adaptation (LoRA) and bias-only optimization (BitFit)—stabilized by a Curriculum Learning strategy that progressively introduces geometrically harder image pairs. For the inference pipeline, we explicitly address background clutter through a Segment-Aware module, leveraging the Segment Anything Model (SAM) to mask irrelevant regions directly within the cost volume. Finally, to mitigate spatial quantization errors, we introduce an Adaptive Window Soft-Argmax that dynamically scales the search radius based on heatmap entropy, achieving fully differentiable sub-pixel accuracy. Evaluated on the SPair-71k benchmark, our approach significantly surpasses the training-free baseline, demonstrating marked improvements in Percentage of Correct Keypoints (PCK), particularly at strict localization thresholds.

1. Introduction

Semantic correspondence aims to establish dense pixel-level matches between images depicting semantically related objects, a task inherently complicated by severe intra-class variations such as drastic changes in viewpoint, scale, and background clutter [8]. Recently, Visual Foundation Models (VFMs), particularly self-supervised Vision Transformers like DINOv2 [9], have demonstrated decent baseline capabilities in dense matching by extracting features that naturally encode rich semantic structures [10, 11]. However, relying solely on training-free frozen features and standard argmax matching presents critical limitations. Generic features lack task-specific adaptation, while standard discrete argmax operations restrict predictions to a rigid grid, introducing quantization errors that prevent sub-pixel accuracy [12].

To overcome these challenges, we present a comprehensive pipeline that adapts and enhances DINOv2 for robust semantic correspondence. To avoid computationally prohibitive full fine-tuning, We first evaluate baseline feature adaptability using Linear Probing, and subsequently explore Parameter-Efficient Fine-Tuning (PEFT) to specialize the backbone. Specifically, we integrate Low-Rank Adaptation (LoRA) [6] to efficiently tune the attention layers, and we systematically compare it against BitFit [2], a sparse optimization technique that updates only the network's bias parameters. This comparative approach allows us to demonstrate how much semantic shift can be resolved by minimal coordinate translations. To stabilize optimization, we introduce a Curriculum Learning strategy [3] exposing the model progressively to increasingly challenging image pairs based on scale, truncation, and viewpoint variations. . At the prediction level, we replace the conventional discrete matching rule with an entropy-driven Soft-Argmax adaptive window [12]. By computing the Shannon entropy of the similarity distribution, our method dynamically modulates the search radius, enabling precise sub-pixel localization for confident matches while encouraging broader exploration under uncertainty. Finally, to explicitly suppress background distractors, we propose Segment Anything Model (SAM) [7] to constrain the cost volume strictly within the target object's boundaries.

In summary, our main contributions are as follows:

- An extensive comparative evaluation of network adap-

tation strategies, contrasting a frozen Linear Probing baseline with Parameter-Efficient Fine-Tuning (PEFT) approaches—specifically LoRA [6] and BitFit [2]—to achieve high-precision matching with a minimal parameter footprint.
- A Curriculum Learning sampling strategy [3] that ranks training pairs by annotation difficulty (scale, viewpoint, and truncation) to optimize and analyze the training dynamics of both probing and fine-tuning pipelines.
- A novel Adaptive Window Soft-Argmax formulation [12] that leverages heatmap entropy for robust sub-pixel refinement.
- A Segment-Aware matching mechanism using SAM-generated masks [7] to eliminate false positives in cluttered environments.

2. Related work

2.1. Semantic Correspondence

The goal of semantic correspondence is to find dense, pixel-level matches between images containing objects of the same category, overcoming significant intra-class variations. Early approaches heavily relied on hand-crafted descriptors and geometric regularizations. With the advent of deep learning, methods shifted toward using Convolutional Neural Networks (CNNs) to extract dense feature maps, optimizing correlation volumes to establish matches. To rigorously evaluate these methods, large-scale benchmarks such as SPair-71k [8] were introduced, providing high-quality keypoint annotations and rich metadata (e.g., viewpoint and scale variations) to assess robustness under challenging conditions. In our work, we extensively utilize SPair-71k not only for evaluation but also to inform our training dynamics through its geometric metadata.

2.2. Visual Foundation Models

Recently, Visual Foundation Models (VFMs) trained on massive datasets have revolutionized various computer vision tasks. In particular, self-supervised Vision Transformers like DINOv2 [9] have been shown to produce feature representations that natively encode strong semantic and spatial structures. Subsequent studies, such as DIFT [10] and the work by Zhang et al. [11], demonstrated that these emergent properties can be directly exploited to perform training-free semantic correspondence without any task-specific training. While these training-free baselines achieve impressive results, they are often sensitive to background noise and lack the precision required for fine-grained keypoint matching. To mitigate the impact of cluttered scenes, we complement the DINOv2 backbone with the Segment Anything Model (SAM) [7], exclusively at inference time to mask irrelevant cost volume regions. This training-free, plug-and-play enhancement avoids training overheads while consistently boosting performance by 0.5%–2.0% across all evaluated architectures (baseline, Linear Probe, and PEFT).

2.3. Linear Probing vs. Fine-Tuning

When adapting Foundation Models, linear probing trains only a task-specific head (e.g., an MLP) while keeping the feature extractor frozen, whereas full fine-tuning updates all network parameters, risking catastrophic forgetting. To evaluate baseline adaptability, we implemented a Linear Probe strategy with an Adaptive Window (AW) mechanism. This approach achieves 65.69% PCK@0.10 on the SPair-71k dataset—a significant improvement over the DINOv2 zero-shot baseline (43.88%)—while adding just 0.263M trainable parameters. Additionally, we explored Curriculum Learning to optimize the training. Although it did not improve final accuracy (yielding 60.56% PCK@0.10), it successfully halved the overall training time, providing a valuable trade-off when computational efficiency is prioritized.

2.4. Parameter-Efficient Fine-Tuning

While full fine-tuning offers theoretical advantages, it is computationally prohibitive and prone to the aforementioned catastrophic forgetting. To address this, Parameter-Efficient Fine-Tuning (PEFT) techniques have been widely adopted. Low-Rank Adaptation (LoRA) [6], originally proposed for Large Language Models, has proven highly effective in vision tasks by injecting trainable low-rank decomposition matrices into the transformer's attention layers while keeping the original weights frozen. Alongside LoRA, sparse optimization methods such as BitFit [2] have demonstrated that competitive performance can often be achieved by updating exclusively the network's bias terms, leaving all weight matrices completely frozen. In our work, we adopt and explicitly compare both LoRA and BitFit. This comparative approach not only allows us to efficiently adapt DINOv2 features for semantic correspondence with a negligible parameter footprint, but also enables us to systematically investigate the extent to which complex spatial matching phenomena can be resolved through simple coordinate translations in the feature space.

2.5. Matching Refinements and Training Strategies

Standard correspondence pipelines typically compute a cosine similarity cost volume followed by a discrete argmax operation to localize the best match. However, as highlighted by Zhang et al. [12], this discrete operation restricts predictions to the rigid resolution of the feature grid, introducing severe quantization errors. To achieve sub-pixel accuracy, they proposed a fixed-window soft-argmax refinement [12]. We build upon this concept by introducing an *entropy-driven* Adaptive Window Soft-Argmax, which dynamically scales the refinement radius based on the confidence (Shannon entropy) of the similarity heatmap. Finally, recognizing that image pairs in semantic matching exhibit substancial variance in difficulty, we draw inspiration from Curriculum Learning [3]. By sorting training samples based on scale, truncation, and viewpoint variations, we gradually expose the model to harder examples, ensuring a more stable and effective optimization process.

3. Method and Workflow

3.1. Feature Extraction and Cost Volume

Given an image pair (I_s, I_t), the first step in our framework is the extraction of dense semantic representations. We employ a Vision Transformer backbone, specifically DINOv2 [9], which natively encodes rich patch-level features. The raw features extracted from a deep intermediate layer are optionally passed through a lightweight projection head and subsequently ℓ_2-normalized. This yields dense feature maps $\mathbf{F}^s \in \mathbb{R}^{N_s \times C}$ and $\mathbf{F}^t \in \mathbb{R}^{N_t \times C}$, where $N_s = N_t = h \times w$ represents the flattened spatial resolution of the feature grid, and C denotes the channel dimensionality.

To evaluate the semantic similarity between spatial locations, we compute the pair-wise cosine similarity between all source and target feature vectors via batched matrix multiplication. The resulting cost volume $\mathbf{S} \in \mathbb{R}^{N_s \times N_t}$ is formulated as:

$$\mathbf{S}(i, j) = \frac{\langle \mathbf{F}_i^s, \mathbf{F}_j^t \rangle}{\tau_{cv}} \tag{1}$$

where $\mathbf{F}_i^s$ and $\mathbf{F}_j^t$ represent the i-th and j-th feature vectors of the source and target images respectively, and τ_{cv} is an initial temperature scaling factor, empirically set to 0.05, that strictly controls the initial sharpness of the cosine similarity distribution prior to any geometric refinement.

In a standard training-free setup, the corresponding target keypoint for a given source query i is obtained by applying a discrete $\arg\max$ operation over the target spatial dimensions j of $\mathbf{S}$. However, this foundational step serves only as the baseline upon which we build our parameter-efficient adaptation and geometric refinements.

3.2. Parameter-Efficient Adaptation: LoRA and BitFit

To adapt the DINOv2 feature space for the precise spatial requirements of semantic correspondence, we implement a progression of network adaptation strategies.

First, we establish a Linear Probing baseline, which leaves the DINOv2 backbone completely frozen and trains only a lightweight projection head (MLP) on top of the feature grid. We then investigate two distinct Parameter-Efficient Fine-Tuning (PEFT) strategies—Low-Rank Adaptation (LoRA) [6] and BitFit [2]—where the backbone adaptation weights are optimized jointly with the projection head.

Low-Rank Adaptation (LoRA) is based on the hypothesis that the weight updates during adaptation lie within a space of low intrinsic rank. Rather than modifying the full pre-trained weight matrix $\mathbf{W} \in \mathbb{R}^{d_{out} \times d_{in}}$, LoRA keeps $\mathbf{W}$ frozen and parameterizes the update $\Delta\mathbf{W}$ using two low-rank matrices $\mathbf{A} \in \mathbb{R}^{r \times d_{in}}$ and $\mathbf{B} \in \mathbb{R}^{d_{out} \times r}$, where the rank satisfies $r \ll \min(d_{in}, d_{out})$. For a given input vector $\mathbf{x}$, the modified forward pass is formulated as:

$$\mathbf{h} = \mathbf{W}\mathbf{x} + \Delta\mathbf{W}\mathbf{x} = \mathbf{W}\mathbf{x} + \frac{\alpha}{r}\mathbf{B}\mathbf{A}\mathbf{x} \tag{2}$$

where α is a constant scaling factor. During initialization, $\mathbf{A}$ is drawn from a random Gaussian distribution while $\mathbf{B}$ is initialized to zero, ensuring $\Delta\mathbf{W} = 0$ at the start of training. In our architecture, these trainable bypasses are specifically integrated into the attention layers (query, key, value, and output projections) of the transformer blocks.

To study an even sparser parameter-efficient strategy, we contrast LoRA with BitFit [2]. This baseline assumes that the task-specific semantic shift can be resolved primarily through coordinate translations in the feature space. In this setup, all weight matrices $\mathbf{W}$ are frozen, and only the bias terms are updated:

$$\mathbf{h} = \mathbf{W}\mathbf{x} + (\mathbf{b} + \Delta\mathbf{b}) \tag{3}$$

where $\mathbf{b}$ is the pre-trained bias and $\Delta\mathbf{b}$ represents the trainable bias update. Programmatically, we implement this by unfreezing all existing bias parameters across the transformer structure.

3.3. Curriculum Learning

Datasets for semantic correspondence, such as SPair-71k [8], exhibit extreme variance in pair difficulty. Uniformly training on this heavy-tailed distribution from the first epoch often induces gradient instability. To address this, we implement a Curriculum Learning [3] strategy to progressively expose the model to increasingly challenging transformations.

Instead of relying on arbitrary heuristics, we compute a normalized difficulty score $d(I_s, I_t) \in [0, 1]$ for each training pair using explicit dataset metadata: viewpoint ($v \in [0, 3]$), scale ($s \in [0, 3]$), and truncation ($t \in [0, 3]$) variations:

$$d(I_s, I_t) = \text{clip}\left(\frac{0.5v + 0.3s + 0.2t}{3.0},\ 0,\ 1\right) \tag{4}$$

Training pairs are sorted in ascending order of difficulty. We then define a linear pacing function $\rho(e)$ that dictates the proportion of the dataset available at training epoch e. Starting from an initial fraction $\rho_{start} = 0.3$ to stabilize early weights, the difficulty ramps up over $E_{ramp} = 10$ epochs:

$$\rho(e) = \min\left(1.0,\ \rho_{start} + \frac{e-1}{E_{ramp}-1}(1-\rho_{start})\right) \tag{5}$$

For any epoch $e \geq E_{ramp}$, the model is exposed to the unconstrained dataset ($\rho(e) = 1.0$). This pacing strategy allows the model to securely anchor basic semantic alignments on simple pairs before tackling severe geometric transformations, effectively mitigating gradient explosions and smoothing the optimization trajectory.

3.4. Inference Pipeline: Segment-Aware Sub-Pixel Refinement

While the adapted DINOv2 features provide robust semantic representations, standard matching during inference still suffers from global false positives induced by background clutter, and local quantization errors caused by discrete $\arg\max$ operations. To resolve both limitations, we introduce a dual-stage, training-free inference refinement pipeline.

Global Segment-Aware Masking. To suppress background distractors, we leverage the Segment Anything Model (SAM) [7] as a zero-shot segmenter. Given a target image, we extract its binary object mask and downsample it via 2D adaptive average pooling to the feature grid resolution $h \times w$. Using an empirical threshold $\theta_{mask} = 0.3$, we obtain a boolean mask $\mathbf{M}_{grid}$. We then apply a hard constraint to the original similarity matrix $\mathbf{S}$:

$$\tilde{\mathbf{S}}(i,j) = \begin{cases} \mathbf{S}(i,j) & \text{if } \mathbf{M}_{grid}(j) = 1 \\ -\infty & \text{otherwise} \end{cases} \tag{6}$$

Setting the similarity of background regions to negative infinity guarantees their softmax probability drops to exactly zero, effectively immunizing the model against out-of-context matches.

Local Adaptive Window Soft-Argmax. Having isolated the object globally, we address local sub-pixel accuracy. Standard discrete $\arg\max$ inherently restricts coordinates to the centers of coarse feature patches. Building upon [12], we introduce an entropy-driven adaptive window mechanism.

For a source keypoint i, we compute the normalized Shannon entropy $H_i \in [0,1]$ of the masked probability distribution. We scale the search window radius R_i based on H_i via piecewise linear interpolation between $R_{min} = 2$ and $R_{max} = 7$, bounded by confidence thresholds $t_{low} = 0.2$ and $t_{high} = 0.7$. Highly confident peaks (low entropy) shrink the window for precise localization, while uncertain matches expand it to smooth over noisy activations.

Given the discrete peak coordinate $\mathbf{y}^* = \arg\max \tilde{\mathbf{S}}$, we define a local window $\mathcal{W}(R_i)$ centered around $\mathbf{y}^*$. The continuous sub-pixel target coordinate $\hat{\mathbf{y}}_i$ is computed as the spatial expectation within this window:

$$\hat{\mathbf{y}}_i = \sum_{\mathbf{y} \in \mathcal{W}(R_i)} \mathbf{y} \cdot \frac{\exp(\tilde{\mathbf{S}}_i(\mathbf{y})/\tau_{win})}{\sum_{\mathbf{y}' \in \mathcal{W}(R_i)} \exp(\tilde{\mathbf{S}}_i(\mathbf{y}')/\tau_{win})} \tag{7}$$

where $\tau_{win} = 0.02$ is a local temperature scaling factor. This sequential combination seamlessly guarantees matching strictly within the target object while dynamically balancing sub-pixel precision and robustness.

3.5. Training Objective

To explicitly optimize the spatial localization capabilities of our LoRA-adapted backbone, we train the network end-to-end using a fully supervised coordinate regression objective. Leveraging the differentiability of our Adaptive Window Soft-Argmax formulation, we compute the continuous Mean Squared Error (MSE) between the predicted sub-pixel coordinates $\hat{\mathbf{y}}_i$ and the ground-truth annotations $\mathbf{y}_i^{gt}$.

Since image pairs contain a variable number of valid matching points due to occlusions or truncation, we apply a boolean validity mask to ensure the loss is exclusively computed over visible keypoints. For a given batch, let $\mathcal{V}$ denote the set of all valid source keypoints across all image pairs. The overall training objective is defined as:

$$\mathcal{L}_{mse} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} ||\hat{\mathbf{y}}_i - \mathbf{y}_i^{gt}||_2^2 \tag{8}$$

This spatial regression loss backpropagates directly through the expectation operation of the soft-argmax, explicitly guiding the trainable low-rank attention matrices to refine the alignment of the dense semantic feature maps.

4. Experimental Setup

4.1. Datasets and Preprocessing

To train and evaluate our models comprehensively, we utilize two standard benchmarks in semantic correspondence: SPair-71k [8] and PF-Pascal.

SPair-71k. The SPair-71k dataset serves as our primary large-scale benchmark for both parameter-efficient fine-tuning and evaluation. It comprises 70,958 annotated image pairs distributed across 18 diverse object categories. SPair-71k provides rich metadata for each image pair, including explicit annotations for viewpoint variation, scale variation, and truncation. We leverage these annotations to objectively quantify the geometric difficulty of each pair.

PF-Pascal. To evaluate the zero-shot cross-dataset generalization capabilities of our pipeline, we additionally employ the PF-Pascal dataset. By evaluating the model on PF-Pascal without any dataset-specific fine-tuning, we verify that our LoRA adaptations and geometric refinements (Soft-Argmax and Segment-Aware masking) do not overfit to the

SPair-71k distribution but learn robust, generalized semantic matching principles.

Preprocessing. Consistent with the standard requirements for Vision Transformer backbones, all input source and target images are resized to a fixed spatial resolution of 224×224 pixels. Subsequently, the images are converted to tensors and normalized using standard ImageNet statistics (mean $\mu = [0.485, 0.456, 0.406]$ and standard deviation $\sigma = [0.229, 0.224, 0.225]$). During evaluation, predicted keypoint coordinates mapped on the 224×224 space are re-scaled back to the original image dimensions to compute the evaluation metrics accurately. Furthermore, to construct uniform batches across variable numbers of annotated keypoints per pair, we dynamically pad the keypoint tensors with dummy values (-1.0) and utilize boolean masks to ensure metrics are computed strictly over valid annotations.

4.2. Evaluation Metric

Following standard practices in semantic correspondence literature, we measure model performance using the Percentage of Correct Keypoints (PCK) [8]. The PCK metric computes the ratio of predicted keypoints that fall within a defined tolerance radius from the ground-truth annotations. To account for scale variations across different object sizes, this tolerance is dynamically scaled based on the object's bounding box rather than the absolute image dimensions.

Furthermore, since image pairs in SPair-71k contain a variable number of annotated keypoints, we pad the tensors to construct uniform GPU batches and employ a boolean mask to ensure the metric is computed exclusively over valid keypoints. Formally, for a set of valid keypoints $\mathcal{V}$, the PCK at a specific threshold α is defined as:

$$\text{PCK@}\alpha = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \mathbf{1}\left[\|\hat{\mathbf{p}}_i - \mathbf{p}_i\| \leq \alpha \cdot \max(H_{bbox}, W_{bbox})\right] \quad (9)$$

where $\hat{\mathbf{p}}_i$ and $\mathbf{p}_i$ represent the predicted and ground-truth coordinates of the i-th keypoint, and $\max(H_{bbox}, W_{bbox})$ denotes the maximum dimension of the object's bounding box.

To provide a comprehensive analysis of our pipeline's localization accuracy, we report the PCK at two strict thresholds: $\alpha = 0.1$ and $\alpha = 0.05$. The latter, in particular, requires highly precise sub-pixel matching and serves as a robust indicator of the improvements yielded by our geometric refinements. We evaluate these metrics both globally and on a per-category basis to highlight performance across different object types.

5. Experiments

5.1. Implementation Details

All experiments are implemented in PyTorch and executed on the Google Colab platform, utilizing NVIDIA Tesla T4 and L4 GPUs to leverage scalable computational resources.

We adopt the ViT-Base variant of DINOv2 (ViT-B/14) as our primary feature extractor. During parameter-efficient fine-tuning, the backbone's pre-trained weights are strictly frozen. For our LoRA adaptation, we inject trainable low-rank matrices specifically into the attention blocks (query, key, value, and output projections), setting the rank $r = 16$ and the scaling factor $\alpha = 32$. To highlight the efficiency of our approach, this LoRA implementation requires optimizing only **0.906M** parameters (representing just **1.036%** of the backbone). Alternatively, for our BitFit baseline, we unfreeze only the bias parameters across the network, optimizing a mere **0.082M** parameters (**0.093%**).

The models are optimized using the AdamW optimizer with an initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-2}. We utilize a Cosine Annealing learning rate scheduler over a total of 20 epochs. Training is conducted with a batch size of 16 image pairs per iteration, leveraging Automatic Mixed Precision (AMP) to accelerate convergence while managing the memory footprint.

For the Curriculum Learning strategy, the pacing function is initialized with $\rho_{start} = 0.3$, linearly ramping up to the full dataset over $E_{ramp} = 10$ epochs. For the Linear Probe baseline and feature projection, we map the extracted features to a dimensionality of 256 and compute the initial cost volume using a temperature scaling factor of $\tau_{cv} = 0.05$. Finally, for the Adaptive Window Soft-Argmax, the entropy thresholds are bounded by $t_{low} = 0.2$ and $t_{high} = 0.7$, dynamically scaling the search radius between $R_{min} = 2$ and $R_{max} = 7$ feature patches. The local sub-pixel expectation temperature is fixed at $\tau_{win} = 0.02$.

5.2. Ablation Studies and Calibration

Before evaluating the end-to-end performance of our pipeline, we ablate key hyper-parameters and architectural choices to empirically justify our experimental setup. We evaluate these preliminary ablations on the SPair-71k validation set using the strict PCK@0.1 metric.

Layer Selection. Vision Transformers encode different hierarchical features across their depth. Early layers typically capture low-level textures and edges, while deeper layers encode high-level semantic concepts. We evaluate the zero-shot matching performance of pure DINOv2 features extracted from different depths (Layer 4, Layer 8, and the final Layer 11). As shown in Table 1, early layers yield poor localization (**13.96%** for Layer 4), while the final Layer 11 achieves the most robust semantic representations (**43.38%**), justifying our choice to extract features from the deepest layer.

Temperature Calibration. The sharpness of the initial cosine similarity distribution directly impacts the behavior of the subsequent Softmax operations and entropy calcu-

Table 1. Layer-wise zero-shot performance on SPair-71k validation set.

Feature Source (DINOv2)	PCK@0.1 (%)
Layer 4	13.96
Layer 8	20.15
Layer 11 (Default)	**43.38**

lations. We ablate the initial cost volume temperature τ_{cv} across multiple magnitudes. Table 2 demonstrates that an overly sharp distribution ($\tau_{cv} = \mathbf{0.01}$) causes premature peak collapse, while a flat distribution ($\tau_{cv} = \mathbf{0.5}$) obscures the optimal match, plummeting accuracy to **26.32%**. Best performance is achieved at $\tau_{cv} = \mathbf{0.05}$.

Table 2. Effect of initial cost volume temperature scaling on matching accuracy.

Temperature (τ_{cv})	PCK@0.1 (%)
0.01	80.98
0.05	**81.87**
0.10	79.00
0.50	26.32

Impact of Curriculum Learning. While evaluating the convergence of our LoRA adaptation, we observe an interesting trade-off introduced by the Curriculum Learning strategy. As detailed in Table 3, standard training yields a marginally higher peak PCK across both configurations (reaching **80.53%** with the Adaptive Window, compared to **79.60%** under the curriculum regime). However, this minor degradation in absolute accuracy is strongly offset by a dramatic reduction in computational cost: the curriculum strategy effectively halves the overall training time. By processing only a progressive fraction of the dataset during the ramp-up phase, the curriculum acts as a robust regularizer—preventing early gradient spikes and ensuring optimization stability—while simultaneously doubling the overall training speed.

Table 3. Impact of Curriculum Learning on SPair-71k matching performance (PCK@0.1).

Method	Standard Training	Curriculum Learning
Linear Probe No AW	65.69%	60.56%
Linear Probe + AW	66.47%	60.89%
LoRA No AW	78.13%	77.13%
LoRA + AW	**80.53%**	79.60%

Architectural Ablation. To isolate the contribution of our proposed modules, we evaluate them against the training-free DINOv2 and DINOv3 baselines. Table 4 illustrates that while standard LoRA significantly improves upon the baseline (reaching **78.13%**), the integration of our Adaptive Window Soft-Argmax systematically mitigates quantization errors, pushing the accuracy to its peak (**80.53%**).

Table 4. Component-wise ablation and baseline comparison on the SPair-71k dataset.

Method	PEFT	Soft-Argmax	PCK@0.1 (%)
Baseline DINOv2	None	×	43.88
Baseline DINOv3	None	×	44.49
SAM (ViT-B)	None	×	6.41
BitFit No AW	BitFit	×	75.40
BitFit + AW	BitFit	✓	77.83
LoRA No AW	LoRA	×	78.13
LoRA + AW	LoRA	✓	**80.53**

5.3. Main Results on SPair-71k

We compare our fully equipped pipeline (DINOv2 + LoRA + Adaptive Window) against the training-free baseline on the SPair-71k test split. To rigorously evaluate the sub-pixel precision of our approach, we report performance at both the standard threshold ($\alpha = \mathbf{0.1}$) and a strict localization threshold ($\alpha = \mathbf{0.05}$).

As detailed in Table 5, our proposed method achieves state-of-the-art performance within the DINOv2 framework. At the standard PCK@0.1 threshold, our model reaches a performance of **80.53%**, an improvement of **+36.65%** over the frozen baseline (**43.88%**). The impact of our Adaptive Window Soft-Argmax is particularly evident at the strict PCK@0.05 threshold: our method maintains a **61.36%** accuracy, whereas the discrete argmax baseline collapses to **24.28%**. This confirms that our entropy-driven refinement effectively solves the structural quantization error inherent to rigid ViT feature grids.

Table 5. Quantitative comparison on the SPair-71k dataset.

Method	PCK@0.1	PCK@0.05
DINOv2 Baseline [9]	43.88%	24.28%
LoRA + AW	**80.53%**	**61.36%**

5.4. Exploratory Project Extensions

Cross-Dataset Generalization. A critical risk of parameter-efficient fine-tuning is overfitting to the specific appearance distribution of the training dataset. To verify the generalization capabilities of our learned spatial representations, we evaluated the model—trained strictly on SPair-71k—directly on the PF-Pascal dataset in a zero-shot setting. As shown in Table 6, the zero-shot DINOv2 baseline achieves **26.76%** PCK@0.1 on PF-Pascal, with DINOv3 showing identical top-level performance. Our parameter-efficient adaptations generalize to the new dataset without any further fine-tuning: BitFit reaches **46.26%**, and our LoRA-adapted model elevates the performance to **47.92%**. This confirms that our training pipeline learns highly robust semantic matching principles rather than merely memorizing SPair-71k instances.

Table 6. Cross-Dataset Generalization: Training (SPair) vs Evaluation (Pascal).

Method	Training (SPair) PCK@0.1	Evaluation (Pascal) PCK@0.1	Evaluation (Pascal) PCK@0.05
DINOv2 Baseline	43.88%	26.76%	14.12%
DINOv3 Baseline	44.49%	26.76%	14.30%
BitFit (Finetuned)	77.83%	46.26%	31.57%
LoRA + AW	**80.53%**	**47.92%**	**35.02%**

Figure 1. Baseline vs. Trained Model: while the training-free baseline mislocalizes the query point to the bow, our fine-tuned pipeline accurately identifies it on the midships.

SAM as an Alternative Baseline. We initially considered the Segment Anything Model (SAM) [7] not only as a masking tool but as a primary feature extractor for the correspondence task. As a prominent Vision Foundation Model, the SAM encoder represents a potential alternative to DINOv2. However, our preliminary evaluations in a training-free setting revealed a significant performance gap. While the DINOv2 baseline achieved a **PCK@0.1** of **43.88%** on SPair-71k, the SAM-based feature matching yielded only **6.41%**. This confirms that SAM's internal representations are highly specialized for object boundary identification rather than for semantic part-alignment. Although the project documentation mentions applying transfer learning to alternative backbones, we decided to focus our computational resources on adapting DINOv2 via LoRA. Fine-tuning the massive SAM encoder would have introduced a prohibitive computational overhead without guaranteeing a superior semantic prior. Consequently, we opted to utilize SAM exclusively for its superior zero-shot segmentation capabilities within our Segment-Aware masking module.

Robustness to Geometric Transformations. Semantic correspondence in the wild requires resilience against severe viewpoint and pose variations. We explicitly stress-test our model by synthetically rotating the target images of the SPair-71k dataset by $\mathbf{45^\circ}$, $\mathbf{90^\circ}$, and $\mathbf{180^\circ}$.

Table 7. Robustness to geometric transformations. PCK@0.1 (%) across different synthetic rotation angles on the SPair-71k dataset.

Rotation Angle	Baseline DINOv2	LoRA + AW
0° (No Rotation)	43.88%	**80.53%**
45°	35.63%	**61.60%**
90°	27.55%	**42.60%**
180°	22.82%	**28.06%**

Figure 2. Robustness to severe geometric transformations and intra-class variance. The model successfully matches the foot of a seagull to an emu in an upright position. Even under a complete 180° inversion of the target image, our LoRA-adapted model retains the spatial prior and correctly localizes the semantic part.

As detailed in Table 7, while arbitrary rotations inherently degrade the spatial priors of Vision Transformers, our adapted model demonstrates superior robustness across all distortion levels. Under intermediate rotations (45° and 90°), our approach maintains a commanding lead over the training-free baseline. Even under a complete 180° inversion, LoRA continues to outperform the baseline, resisting the failure typically observed in frozen features. This empirical evidence highlights the efficacy of combining task-specific LoRA attention patterns with dynamic search windows to recover correspondences even under severe geometric distortions.

6. Conclusion

In this paper, we presented a modular pipeline to adapt Visual Foundation Models for high-precision semantic correspondence, addressing the lack of task-specific tuning in training-free Vision Transformers. Our exploration of PEFT strategies demonstrated that LoRA achieves peak performance, while a Curriculum Learning strategy significantly accelerates convergence by halving probing training time. Furthermore, integrating our entropy-driven Adaptive Window Soft-Argmax and Segment-Aware SAM module consistently refined sub-pixel accuracy across all configurations on the SPair-71k benchmark. Finally, downstream evaluations validated the robust generalization of our optimized architecture, demonstrating substantial improvements on zero-shot cross-dataset transfer to PF-Pascal and strong resilience to severe geometric rotations.

6.1. Limitations and Future Work

Despite strong empirical performance, our approach has limitations that open avenues for future research. First, the Segment-Aware module inherently bounds the pipeline's upper limit to SAM's zero-shot capabilities: severe truncation or motion blur can cause inaccurate masks that inadvertently suppress valid semantic regions. To resolve this, future work will explore upgrading the post-processing stage to the newly released **Meta (Facebook) SAM 3** [4], leveraging its promptable concept segmentation to dynamically refine keypoint boundaries. Second, maintaining high-resolution feature grids (e.g., 14×14 patch size) with attention tuning incurs a non-negligible computational footprint, potentially limiting real-time edge applications.

To address these challenges, we plan to expand this architecture to temporal domains, enforcing spatio-temporal consistency for dense video correspondence. Furthermore, we will systematically benchmark alternative self-supervised paradigms within the **Meta (Facebook) AI** foundation ecosystem. Specifically, evaluating **Meta's Masked Autoencoder (MAE)** [5] will allow us to isolate the exact empirical benefits of reconstructive vs. self-distillative objectives. Additionally, transitioning to **Meta's Image Joint Embedding Predictive Architecture (I-JEPA)** [1] represents a highly promising direction; by predicting masked patches in the *representation space* rather than pixel space, I-JEPA discards high-frequency background noise to focus entirely on the high-level structural features critical for robust spatial matching.

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